

Post Graduate Professional Program (PGPP) in Artificial Intelligence & Machine Learning

Duration: 10 Months (40 Weeks)

Mode: Online / Offline

Classes: 3 Days per Week

Class Duration: 1.5 Hours

Target Audience: Fresh Graduates, Final Year Students, Working Professionals

Program Objective

By the end of the program, students should be able to:

- Analyze real-world datasets
- Build Machine Learning models
- Develop Deep Learning applications
- Create Generative AI applications
- Deploy AI solutions to the cloud
- Work independently on industry projects
- Build a strong GitHub portfolio

Phase 1: Programming & Data Foundation (Month 1-2)

Module 1: Python for AI & ML

Topics

- Python Fundamentals
- Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- OOP
- Exception Handling
- File Handling
- Modules & Packages

Practical Work

- Student Management System
- Employee Payroll System
- ATM Simulation
- Inventory Management System

Assignment

- Build a Console-Based Library Management System

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Module 2: SQL for Data Professionals

Topics

- Database Fundamentals
- MySQL
- CRUD Operations
- Joins
- Group By
- Views
- Stored Procedures
- Window Functions

Practical Work

- E-commerce Database
- Employee Database
- Hospital Database

Industry Case Study

- Analyze customer purchase data using SQL

Module 3: Git & GitHub

Topics

- Version Control
- Branching
- Pull Requests
- Collaboration Workflow

Practical

- Host all assignments on GitHub

Phase 2: Data Science & Analytics (Month 3-4)

Module 4: Data Analysis using Pandas & NumPy

Topics

- Data Cleaning
- Data Transformation
- Missing Values
- Feature Engineering

Practical Datasets

- Netflix Dataset
- IPL Dataset
- E-Commerce Dataset

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Module 5: Data Visualization

Tools

- Matplotlib
- Seaborn
- Plotly

Practical Work

- Business Sales Dashboard
- Customer Analytics Dashboard

Module 6: Statistics for AI

Topics

- Mean, Median, Mode
- Variance
- Standard Deviation
- Probability
- Hypothesis Testing
- Correlation
- Regression

Practical

- Analyze business datasets statistically

Phase 3: Machine Learning Engineering (Month 5-6)

Module 7: Machine Learning Fundamentals

Topics

- ML Lifecycle
- Feature Selection
- Train/Test Split
- Cross Validation
- Evaluation Metrics

Algorithms

Regression

- Linear Regression
- Polynomial Regression

Classification

- Logistic Regression
- KNN
- Decision Trees
- Random Forest

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- Naive Bayes
- SVM

Clustering

- K-Means
- Hierarchical Clustering

Projects

- House Price Prediction
- Employee Salary Prediction
- Customer Churn Prediction

Module 8: Advanced Machine Learning

Topics

- Ensemble Learning
- XGBoost
- LightGBM
- CatBoost
- Hyperparameter Tuning

Project

- Loan Approval Prediction System

Phase 4: Deep Learning & Computer Vision (Month 7)

Module 9: Deep Learning

Topics

- Neural Networks
- TensorFlow
- Keras
- Backpropagation
- Activation Functions

Practical

- Build Neural Networks from Scratch

Module 10: Computer Vision

Topics

- Image Processing
- CNN
- Transfer Learning
- Object Detection

Projects

- Face Mask Detection
- Vehicle Detection
- Crop Disease Detection

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Phase 5: NLP & Generative AI (Month 8)

Module 11: Natural Language Processing

Topics

- Text Processing
- Tokenization
- Stemming
- Lemmatization
- Sentiment Analysis

Projects

- Movie Review Sentiment Analyzer
- Spam Email Detector

Module 12: Generative AI

Topics

- LLM Fundamentals
- Prompt Engineering
- OpenAI APIs
- Hugging Face
- AI Agents

Practical

- AI Content Generator
- AI Resume Builder
- AI Email Generator

Phase 6: LangChain, RAG & AI Agents (Month 9)

Module 13: Advanced Generative AI

Topics

- LangChain
- LangGraph
- Vector Databases
- ChromaDB
- Pinecone
- RAG Architecture
- Multi-Agent Systems

Projects

- PDF Chatbot
- Company Knowledge Assistant
- HR Policy Chatbot
- College Admission Assistant

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Module 14: AI Product Development

Topics

- FastAPI
- Streamlit
- Gradio

Project

- Deploy AI Applications

Phase 7: MLOps & Industry Deployment (Month 10)

Module 15: MLOps

Topics

- Docker
- CI/CD Basics
- Model Monitoring
- MLflow
- DVC

Module 16: Cloud Deployment

Topics

- AWS Fundamentals
- EC2
- S3
- Render
- Railway
- Vercel

Practical

- Deploy AI Models Live

Capstone Industry Projects (Choose One)

Healthcare

- Medical Report Analyzer
- Disease Prediction System

HR

- Resume Screening AI
- Candidate Ranking System

Education

- AI Learning Assistant
- Student Performance Predictor

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Retail

- Product Recommendation Engine
- Sales Forecasting System

Finance

- Fraud Detection System
- Loan Approval AI

Agriculture

- Crop Disease Detection
- Yield Prediction System

Career Development (Throughout Program)

Technical Preparation

- Python Interview Questions
- SQL Interview Questions
- ML Interview Questions
- Generative AI Interview Questions

Professional Development

- Resume Building
- LinkedIn Optimization
- GitHub Portfolio
- Mock Interviews

Mandatory Portfolio Deliverables

Every student must complete:

- ✓ 25+ Assignments
- ✓ 10+ Mini Projects
- ✓ 5 Major Projects
- ✓ 1 End-to-End ML Project
- ✓ 1 Generative AI Project
- ✓ 1 Live Industry Project
- ✓ GitHub Portfolio
- ✓ Deployed AI Application
- ✓ Professional Resume